

Normalising the Force by $|\nabla\rho|$: Why it Removes the Dip

Abstract

An elongated (elliptical) potential field produces a spurious local minimum of $|F|$ along its major axis — weakest force exactly where the obstacle lies most directly ahead. This note explains, first informally and then with proofs, why dividing the force by $|\nabla\rho|$ removes that minimum for every elongation while leaving the force direction exactly unchanged. The essential observation is that $|\nabla U|$ factors into a part that depends on ρ alone and a part that depends on *direction*, and that the entire defect lives in the second part. We also delimit the result: it is exact on horizontal scans of the implemented field, but the stronger statement that $|F|$ is constant on ellipses fails there, for reasons we make precise.

1 The idea in words

1.1 Force is contour spacing

The magnitude of a gradient is a rate: how much U changes per metre travelled. Equivalently, it measures how tightly the equipotential contours are packed. Where contours crowd, the force is large; where they spread, it is small.

Now elongate the field by a factor q along y . The contours become ellipses stretched $q : 1$. Crucially, *the same potential drop is now spread over q times more metres at the poles of the ellipse than at its flanks*. So the force at the poles is q times weaker — not because the situation there is any safer, but because of how the contours were drawn. The dip is a measurement artefact of this stretching.

1.2 ρ is a risk coordinate, not a length

The cleanest way to see the artefact is to notice that ρ is not a distance. It is a *risk coordinate*: a bookkeeping variable in which one unit longitudinally is worth t_y metres while one unit laterally is worth only t_x metres. When we form ∇U we are implicitly applying the chain rule,

$$\underbrace{\frac{\partial U}{\partial(\text{metres})}}_{\text{what } |\nabla U| \text{ measures}} = \underbrace{\frac{\partial U}{\partial \rho}}_{\text{what we want}} \times \underbrace{\frac{\partial \rho}{\partial(\text{metres})}}_{\text{exchange rate}}, \quad (1)$$

and the trouble is that the exchange rate is *direction-dependent*: it is $1/t_y$ along the major axis and $1/t_x$ along the minor axis. The first factor is the honest quantity — how fast risk changes per unit of risk — and it can have no directional preference, because U is a function of ρ alone. The second factor is pure bookkeeping, and it is the sole source of the dip.

Dividing by $|\nabla\rho|$ cancels the bookkeeping factor and returns $\partial U/\partial\rho$. That is the whole method.

1.3 The eikonal picture

There is a standard name for the property ρ lacks. A genuine distance function satisfies the *eikonal condition*

$$|\nabla\phi| = 1, \quad (2)$$

meaning one metre of travel buys exactly one unit of ϕ , in every direction. Our ρ violates this: its gradient norm ranges over $[1/t_y, 1/t_x]$, a swing of exactly the elongation ratio q . In level-set methods a function that has drifted away from the eikonal condition is repaired by *reinitialisation*; dividing by $|\nabla\rho|$ is the closed-form, one-step version of the same repair. Seen this way, the method is not a trick but the restoration of a property the field was implicitly assumed to have.

1.4 Why the direction survives

Finally, the reason this costs nothing in steering behaviour: $|\nabla\rho|$ is a *positive scalar*. Dividing a vector by a positive scalar changes its length and not its heading. So every arrow in the field keeps pointing exactly where it pointed before; only the colour map changes. This is what distinguishes the method from metric-weighted alternatives such as $-G^{-1}\nabla U$, which rotate the force toward the radial direction and destroy the lateral guidance.

Remark 1 (One-sentence summary). The dip is the gap between “potential drop per metre” and “potential drop per unit of risk”; the normalisation measures the latter, which by construction cannot prefer a direction.

2 Setup and notation

Let the obstacle sit at (x_o, y_o) and write the gaps $d_x = x_o - X$, $d_y = y_o - Y$ for a host at (X, Y) . With per-axis time constants $t_x, t_y > 0$ define the elliptical pseudo-distance

$$\rho = \sqrt{\left(\frac{d_x}{t_x}\right)^2 + \left(\frac{d_y}{t_y}\right)^2}, \quad q = \frac{t_y}{t_x} \text{ (the elongation)}. \quad (3)$$

Throughout, ∇ is taken with respect to the host position (X, Y) ; since d_x, d_y are affine in X, Y with unit slopes, signs cancel in every magnitude below. We consider potentials of the form $U = U(\rho)$ with U' of one sign, and define the normalised force

$$F_n = - \frac{\nabla U}{|\nabla\rho|} \quad (4)$$

3 The mathematics

3.1 Localising the defect

Lemma 2 (Factorisation). *If $U = U(\rho)$ then $\nabla U = U'(\rho) \nabla\rho$, and hence*

$$|\nabla U| = \underbrace{|U'(\rho)|}_{\text{depends on } \rho \text{ only}} \cdot \underbrace{|\nabla\rho|}_{\text{depends on direction}}. \quad (5)$$

Proof. Immediate from the chain rule applied to the composition $U \circ \rho$. □

The two factors have sharply different characters, and the next two results make that precise. Lemma 2 is the formal content of Section 1: the first factor is the honest rate, the second is the exchange rate.

Lemma 3 (Range of the exchange rate). *Away from the obstacle centre,*

$$|\nabla\rho| = \frac{\sqrt{(d_x/t_x^2)^2 + (d_y/t_y^2)^2}}{\rho}, \quad |\nabla\rho| \in \left[\frac{1}{\max(t_x, t_y)}, \frac{1}{\min(t_x, t_y)} \right], \quad (6)$$

with the endpoints attained exactly on the two axes: $|\nabla\rho| = 1/t_y$ on the major axis ($d_x = 0$) and $|\nabla\rho| = 1/t_x$ on the minor axis ($d_y = 0$). The ratio between the extremes is exactly q .

Proof. Differentiating (3) gives $\partial\rho/\partial d_x = d_x/(t_x^2\rho)$ and likewise for d_y , which yields the stated formula. For the range, put $u = (d_x/t_x)^2 \geq 0$ and $v = (d_y/t_y)^2 \geq 0$, so that $\rho^2 = u + v$ and $(d_x/t_x^2)^2 + (d_y/t_y^2)^2 = u/t_x^2 + v/t_y^2$. Hence

$$|\nabla\rho|^2 = \frac{u}{u+v} \cdot \frac{1}{t_x^2} + \frac{v}{u+v} \cdot \frac{1}{t_y^2}, \quad (7)$$

a convex combination of $1/t_x^2$ and $1/t_y^2$ with non-negative weights summing to one. A convex combination lies between its extremes, and attains them precisely when a weight vanishes, i.e. when $v = 0$ or $u = 0$. Taking square roots gives the claim. $\square$

Lemma 3 quantifies the intuition exactly: the offending factor sweeps a range whose width *is* the elongation q . For an isotropic field $t_x = t_y$ the range collapses to a point and there is nothing to correct.

3.2 What the normalisation achieves

Theorem 4 (The normalised magnitude is direction-free). *For $U = U(\rho)$ and F_n as in (4),*

$$|F_n| = |U'(\rho)|, \quad (8)$$

a function of ρ alone. Equivalently, $F_n = -U'(\rho) \hat{n}$ with $\hat{n} = \nabla\rho/|\nabla\rho|$ the outward unit normal to the ellipse through the point.

Proof. By Lemma 2, $|\nabla U| = |U'(\rho)| |\nabla\rho|$. Dividing by $|\nabla\rho| > 0$ gives $|F_n| = |U'(\rho)|$. The vector form follows from $\nabla U = U'(\rho)\nabla\rho$ on dividing by $|\nabla\rho|$. $\square$

Corollary 5 (The two families of level sets coincide). *Under the hypotheses of Theorem 4, the level sets of $|F_n|$ are exactly the level sets of U , namely the ellipses $\rho = \text{const}$.*

Proof. Both $|F_n| = |U'(\rho)|$ and $U = U(\rho)$ are functions of ρ alone, so each is constant on $\{\rho = c\}$; as $|U'|$ is of one sign and ρ ranges over an interval, the sets $\{\rho = c\}$ are precisely the level sets of each. $\square$

Corollary 5 is the sharpest statement of why the method works. The dip was, by definition, a mismatch between the shape of the $|F|$ contours (pinched, non-convex, peanut-like) and the shape of the U contours (nested convex ellipses). The normalisation forces the two families to agree, and a family of nested ellipses has no pinch.

Theorem 6 (No dip, at any elongation). *Fix a height $s > 0$ and suppose $|U'|$ is strictly decreasing in ρ (as for $U = \lambda/\rho$). Then along the horizontal scan $y = s$ the magnitude $|F_n|$ is strictly decreasing in $|x|$, attains its maximum at $x = 0$, and has no interior critical point. This holds for every $q > 0$, with no threshold.*

Proof. By Theorem 4, $|F_n| = |U'(\rho)|$. At fixed s ,

$$\rho(x, s) = \sqrt{\left(\frac{x}{t_x}\right)^2 + \left(\frac{s}{t_y}\right)^2} \quad (9)$$

is strictly increasing in $|x|$, since $\partial\rho/\partial x = x/(t_x^2\rho) > 0$ for $x > 0$. The composition of the strictly decreasing map $\rho \mapsto |U'(\rho)|$ with the strictly increasing map $|x| \mapsto \rho$ is strictly decreasing. A strictly monotone function on $[0, \infty)$ attains its maximum at the left endpoint and has no interior stationary point. No hypothesis on q was used. $\square$

Corollary 7 (Direction is exactly preserved). *F_n is everywhere parallel to $-\nabla U$, with the same orientation.*

Proof. $F_n = (-\nabla U)/|\nabla\rho|$ and $|\nabla\rho| > 0$ away from the obstacle centre. Multiplication by a positive scalar preserves both the line spanned by a vector and its orientation. $\square$

Proposition 8 (Consistency: the isotropic case is a fixed point). *If $t_x = t_y = t$ then $|\nabla\rho| \equiv 1/t$, so F_n differs from $-\nabla U$ by the single global constant t ; if moreover $t = 1$ then $F_n = -\nabla U$ identically.*

Proof. Set $t_x = t_y = t$ in Lemma 3: the interval degenerates to the point $1/t$. $\square$

Proposition 8 is the sanity check that the method is a genuine generalisation: it does nothing precisely in the case that had nothing wrong with it, and its strength scales with the elongation that caused the problem.

4 Scope: the implemented field

The implemented potential is *not* of the form $U(\rho)$. It carries a vehicle-dimension prefactor,

$$U_{\text{impl}} = \frac{\lambda}{|k'|}, \quad k' = c(d_x, d_y) \rho, \quad c = \begin{cases} 1 - w/|d_x| & \text{region 1/3,} \\ 1 - l/|d_y| & \text{region 2/4,} \end{cases} \quad (10)$$

and c is not a function of ρ . Two consequences, and it is worth being precise about which results survive.

Proposition 9 (Theorem 6 is exact on horizontal scans). *Restrict (10) to a horizontal line $Y = \text{const}$ lying in region 2/4. Then d_y is constant along that line, hence so is $c = 1 - l/|d_y|$, and $U_{\text{impl}} = (\lambda/c)\rho^{-1}$ is a function of ρ alone along that line. Theorem 6 therefore applies verbatim, and the contrast satisfies $C = 1$ exactly.*

Proof. Along $Y = \text{const}$ we have $d_y = y_o - Y$ fixed, so c is a constant c_0 and $U_{\text{impl}} = (\lambda/c_0)\rho^{-1}$, to which Theorem 6 applies with $U'(\rho) = -(\lambda/c_0)\rho^{-2}$, strictly decreasing in magnitude. $\square$

Remark 10 (Corollary 5 does *not* survive). Along an ellipse $\rho = \text{const}$ the gap d_y varies, so c varies and $|F_n|$ is not constant there. Measured on the implemented field at $q = 15$, the coefficient of variation of $|F_n|$ along arcs of fixed ρ is 52% to 104% — reduced from 105% to 162% for the unnormalised force, but far from zero. The guarantee delivered in practice is thus Proposition 9 (no dip on any horizontal scan), not the stronger Corollary 5.

5 Numerical verification

All figures below are measured on the implemented field at the current configuration $t_x = 1, t_y = 15$, i.e. $q = 15$.

Y	plain C	ridge x^*	normalised C
36.5	1.926	0.288	1.000
37.5	2.418	0.340	1.000
38.5	2.804	0.390	1.000
39.5	3.113	0.438	1.000

Lemma 3 predicts $|\nabla\rho| \in [1/t_y, 1/t_x] = [0.0667, 1]$; sampled over the plotting window the measured range is $[0.0712, 1.0000]$, the lower end approached as the grid nears the major axis. The analytic expression of Lemma 3 agrees with a central-difference evaluation to 6.4×10^{-10} . Corollary 7 is confirmed to machine precision: the force angle at sample points above and below the obstacle is unchanged to 0.00×10^0 degrees.

6 The cost

F_n is not a gradient field. Writing $F_n = -\Phi \nabla\rho$ with $\Phi = U'(\rho)/|\nabla\rho|$, such a field is conservative only if Φ is a function of ρ alone; but by Lemma 3 the factor $|\nabla\rho|$ varies along a single ellipse $\rho = \text{const}$ whenever $q > 1$, so Φ is not, and $\nabla \times F_n \neq 0$. Consequently U is no longer a Lyapunov function for the normalised force. The magnitude of the violation, and the comparison against alternative remedies, are treated in the companion note *Remedies for the Force-Magnitude Dip in an Inverse-Elliptical Potential Field*.

7 Relation to existing work

The manoeuvre has two well-established relatives. In level-set methods, a function that has lost the eikonal property $|\nabla\phi| = 1$ is restored by *reinitialisation*, classically by evolving $\partial_\tau\phi + \text{sgn}(\phi_0)(|\nabla\phi| - 1) = 0$; (4) is the closed-form one-step analogue. In potential-field control, the fully normalised gradient $-\nabla\gamma/\|\nabla\gamma\|$ is used to decouple the commanded direction from a separately regulated magnitude, and incurs the same loss of conservativeness. Equation (4) sits between the two: it removes only the metric factor $|\nabla\rho|$, retaining the physically meaningful variation $|U'(\rho)|$ that a full normalisation would discard.